

The University of New South Wales

Department of Statistics

School of Mathematics

Multivariate Analysis I

Lecture notes

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This volume of notes is for individual students' use only.

**It is therefore not to be distributed beyond the
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Foreword

These notes do **not** substitute the lectures in Multivariate Analysis for masters students. You are strongly recommended to attend each and every lecture and laboratory hour because the conceptual bases of the discussed modeling methods, as well as some additional derivations and explanations will then be focused on, as will be important portions of pertinent computer output. This volume is therefore not meant to be a substitute for a textbook, computer package manual, or lecture attendance.

We rely on the widely spread and powerful statistical package SAS to perform the actual calculations during the course. These notes are a compilation from several sources and other notes. Some of the sources are listed in your handout. As most closest reference book the following source could be mentioned:

Johnson, R & Wichern, D. *Applied Multivariate Statistical Analysis*. Sixth Edition, Prentice Hall.

By no means can this book be a substitute for the whole set of notes, though.

It is assumed that you are familiar with some basic concepts of linear algebra. These will be summarized at the beginning and will be used essentially in the rest of the course. These concepts include matrix and vector operations, determinants, traces, ranks, projectors, linear equations, inverses, eigenvectors and eigenvalues etc.

I will appreciate if you would let me know about any ways these notes could be further improved.

I Lecture

Matrix Algebra and Random Vectors

1. First we shall mention some general aspects of Multivariate Analysis. Usually, when studying complex phenomena, **many** variables are required. Besides, the process of studying is usually an iterative one with many variables often added or deleted from the study. Multivariate analysis deals with developing methods for better understanding the relationships between the many variables included in the analysis of such complex phenomena.

Difficulties in such a process:

- More data to analyse
- More involved mathematics necessary
- Computer intensive methods involved in the process

Objectives of multivariate methods:

- Data reduction (try to present the phenomenon as simply as possible **but** without sacrificing valuable information). Typical *representative method*: Principal components analysis. Sometimes, this reduction is achieved by introducing a small number of unobservable (latent) variables when trying to explain a large number of observable output variables. Representative methods: *factor analysis* and *covariance structure analysis*.
- Sorting or grouping (creating groups of "similar" objects or variables that in a sense are more closer to each other than to objects outside the group). Try to find reasonable explanation for the existing grouping. *Representative methods*: Factor Analysis, Cluster Analysis, Discriminant Analysis.
- Investigation of dependence among variables (Which sets of variables can be considered as independent and which are "more dependent". How can the dependence be "measured"). *Representative Methods*: Correlation Analysis, Partial Correlations, Canonical Correlations.
- Prediction (a basic practically important goal)- one tries to predict values of one or more variables on the basis of observations of other variables that have been found to influence the former variables. *Representative*: Multivariate Regression.
- Hypothesis testing- can be done to either validate assumptions on basis of which certain analysis (e.g., normality) is being done or to reinforce some prior modeling convictions (e.g., equality of parameters etc.) Hypothesis testing is relevant to the applications of all multivariate methods we will be dealing with.

As a basic **mathematical model** for our analyses in this course the **multivariate normal distribution** will be used. Reasons are: our limited time and the complexity of

other approaches. Although in practice also other distributions are relevant, modeling based on the multivariate normal distribution can still be a very good approximation.

2. Data organization. Assume, we are dealing with $p \geq 1$ variables. The values of these variables are all recorded for each distinct *item*, *individual* or *experimental trial*. Each of these three words will be substituted sometimes by the word **case**. We will use the notation x_{ij} to indicate a particular value of the i th variable that is observed on the j th case. Consequently, n measurements on p variables can be represented in a form of a matrix

$$X = X_{(p \times n)} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1j} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2j} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & x_{i2} & \dots & x_{ij} & \dots & x_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{p1} & x_{p2} & \dots & x_{pj} & \dots & x_{pn} \end{pmatrix} \quad (1.1)$$

The matrix X above contains the data consisting of all the observations on all the variables. This way of representing the data allows easy manipulations to be performed in order to obtain some easy descriptive statistics for each of the variables. For example, the *sample mean* of the second variable is just $\bar{x}_2 = \frac{1}{n} \sum_{j=1}^n x_{2j}$, the *sample variance* of the second variable is just $s_2^2 = \frac{1}{n} \sum_{j=1}^n (x_{2j} - \bar{x}_2)^2$ (Note that for the sample variance we shall sometimes use the divisor of $n - 1$ rather than n and each time this will be differentiated by displaying the appropriate expression).

The *sample covariance* (the simple measure of linear association between variables 1 and 2) is given by $s_{12} = \frac{1}{n} \sum_{j=1}^n (x_{1j} - \bar{x}_1)(x_{2j} - \bar{x}_2)$ and one can understand easily how $s_{ik}, i = 1, 2, \dots, p, k = 1, 2, \dots, p$ can be defined.

Finally, the *sample correlation coefficient* (the measure of linear association between two variables that does not depend on the units of measurement) can be defined. The sample correlation coefficient of the i th and k th variables is defined by $r_{ik} = \frac{s_{ik}}{\sqrt{s_{ii}}\sqrt{s_{kk}}}$. Because of the well-known Cauchy-Bunyakovsky-Schwartz Inequality, $|r_{ik}| \leq 1$ holds. Note also that $r_{ik} = r_{ki}$ for all $i = 1, 2, \dots, p, k = 1, 2, \dots, p$ holds.

It should be repeatedly noted that the sample correlations and covariance are useful only when trying to measure the **linear association** between two variables. Their value is less informative and is misleading in cases of *nonlinear* association. In this case one needs to invoke the *correlation quotient* instead. (The precise definitions and the relationships between the correlation coefficient and the correlation quotient will be given at lecture.)

But because of the fact that covariance and correlation coefficients are routinely calculated and analyzed they are very widely used and provide nice numerical summaries of association when the data do not exhibit obvious nonlinear patterns of association.

The descriptive statistics hat we discussed until now are usually organized into arrays, namely:

- Vector of sample means $\bar{x} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \cdot \\ \cdot \\ \bar{x}_p \end{pmatrix}$

- Matrix of sample variances and covariances

$$S = S_n = \begin{pmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ s_{21} & s_{22} & \dots & s_{2p} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{pmatrix} \quad (1.2)$$

- Matrix of sample correlations

$$R = R_n = \begin{pmatrix} 1 & r_{12} & \dots & r_{1p} \\ r_{21} & 1 & \dots & r_{2p} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ r_{p1} & r_{p2} & \dots & 1 \end{pmatrix} \quad (1.3)$$

3. Initial graphical representation of the data. Some simple characteristics of the data are worth studying before the actual multivariate analysis would begin. In SAS the procedures that are used for this purpose are called **proc means**, **proc plot** and **proc corr**. Please, study their short description in the SAS handout and try to use them for the purpose of:

- drawing scatterplot of the data
- calculating simple univariate descriptive statistics for each variable
- calculating sample correlation and covariance coefficients
- linking multiple two-dimensional scatterplots

4. Matrix Algebra

4.1. Vectors and matrices. As a shorthand notation, we shall be using $\mathbf{X} \in \mathcal{M}_{p,n}$ to indicate that $\mathbf{X}$ is a matrix with p rows and n columns. A notation $\mathbf{X} \in \mathbf{R}^n$ will be used to indicate that $\mathbf{X}$ is a n -dimensional *column* vector. Of course, if $\mathbf{X} \in \mathbf{R}^n$, it also means that $\mathbf{X} \in \mathcal{M}_{n,1}$. *Transposition* will be denoted by $'$. After a transposition, from a matrix $\mathbf{X} \in \mathcal{M}_{p,n}$ we get a new matrix $\mathbf{X}' \in \mathcal{M}_{n,p}$. In particular, from a *column* vector $\mathbf{X} \in \mathbf{R}^n$ we arrive, after a transposition, to a *row* vector $\mathbf{X}' \in \mathcal{M}_{1,n}$. It is well known that multiplication of a matrix (vector) with a scalar means multiplication of each of the elements of the matrix (vector) with that scalar. Also, two matrices (vectors) of

the same dimension can be added (subtracted) and the result is a new matrix (vector) of the same dimension and elements which are the element wise sum (difference) of the elements of the matrices (vectors) to be added (subtracted). The *Euclidean norm* of a

$$\text{vector } \mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix} \in \mathbf{R}^p \text{ is denoted by } \|\mathbf{X}\| \text{ and is defined as } \|\mathbf{X}\| = \sqrt{\sum_{i=1}^p \mathbf{x}_i^2}.$$

The *inner product* or, equivalently, the *scalar product* of two p - dimensional vectors $\mathbf{X}$ and $\mathbf{Y}$ is denoted and defined in the following way:

$$(\mathbf{X}, \mathbf{Y}) = \mathbf{X}'\mathbf{Y} = \sum_{i=1}^p \mathbf{x}_i \mathbf{y}_i \quad (1.4)$$

Obviously, the relation $\|\mathbf{X}\|^2 = (\mathbf{X}, \mathbf{X})$ holds. It is well known that if θ is the angle between two p -dimensional vectors $\mathbf{X}$ and $\mathbf{Y}$ then it also holds

$$(\mathbf{X}, \mathbf{Y}) = \|\mathbf{X}\| \cdot \|\mathbf{Y}\| \cdot \cos(\theta) \quad (1.5)$$

Since $|\cos(\theta)| \leq 1$, we have the inequality

$$|(\mathbf{X}, \mathbf{Y})| \leq \|\mathbf{X}\| \cdot \|\mathbf{Y}\|$$

which is one variant of the *Cauchy-Bunyakovsky-Schwartz* Inequality. Further, if we want to *orthogonally project* the vector $\mathbf{X} \in \mathbf{R}^p$ on the vector $\mathbf{Y} \in \mathbf{R}^p$ then (having in mind the geometric interpretation of orthogonal projection) the result will be: $\frac{\mathbf{X}'\mathbf{Y}}{\mathbf{Y}'\mathbf{Y}}\mathbf{Y}$.

Finally, the rules for matrix multiplication are recalled: if $\mathbf{X} \in \mathcal{M}_{p,k}$ and $\mathbf{Y} \in \mathcal{M}_{k,n}$ (i.e. the number of columns in $\mathbf{X}$ is equal to the number of columns in $\mathbf{Y}$) then the multiplication $\mathbf{X}.\mathbf{Y}$ is possible and the result is a matrix $\mathbf{Z} = \mathbf{X}.\mathbf{Y} \in \mathcal{M}_{p,n}$ with elements

$$z_{i,j}, i = 1, 2, \dots, p, j = 1, 2, \dots, n : z_{i,j} = \sum_{m=1}^k x_{i,m} \cdot y_{m,j} \quad (1.6)$$

i.e. the element in the i th row and j th column of $\mathbf{Z}$ is a scalar product of the i th row of $\mathbf{X}$ and the j th column of $\mathbf{Y}$. Note that the multiplication of matrices is **not commutative** and in general, it is not necessary for $\mathbf{YX}$ to even exist when $\mathbf{XY}$ exists. When the matrices are both square (quadratic) of the same dimension p (i.e. both $\mathbf{X} \in \mathcal{M}_{p,p}$ and $\mathbf{Y} \in \mathcal{M}_{p,p}$) then both $\mathbf{XY}$ and $\mathbf{YX}$ will be defined but would in general **not** give rise to the same result.

The following transposition rule is important to be mentioned (and easy to check): if $\mathbf{X} \in \mathcal{M}_{p,k}$ and $\mathbf{Y} \in \mathcal{M}_{k,n}$ then the product $\mathbf{XY}$ exists and it holds:

$$(\mathbf{XY})' = \mathbf{Y}'\mathbf{X}' \quad (1.7)$$

One should be very careful with transposition though in order to avoid silly mistakes. If $\mathbf{X} \in \mathbf{R}^p$, for example, both $\mathbf{X}'\mathbf{X}$ and $\mathbf{XX}'$ exist. While the former is a scalar, the latter belongs to $\mathcal{M}_{p,p}$!

A square matrix $\mathbf{X} \in \mathcal{M}_{\mathbf{p},\mathbf{p}}$ is called **symmetric** if $x_{i,j} = x_{j,i}, i = 1, 2, \dots, p, j = 1, 2, \dots, p$ holds. For such a matrix, we have $\mathbf{X}' = \mathbf{X}$.

The square matrix $\mathbf{I} = \mathbf{I}_{p \times p} = \delta_{ij}, i = 1, 2, \dots, p, j = 1, 2, \dots, p$ (i.e. ones on the diagonal and zeros outside the diagonal) is called the **identity matrix** (of dimension p). Obviously, when the multiplication is possible then always $\mathbf{XI} = \mathbf{X}$ and $\mathbf{IX} = \mathbf{X}$ holds.

4.2. Inverse matrices. Symmetric, orthogonal matrices, spectral decompositions

4.2.1. Inverse matrices

To any **square** matrix $\mathbf{X} \in \mathcal{M}_{\mathbf{p},\mathbf{p}}$ one can attach a number $|\mathbf{X}| = \det(\mathbf{X})$ called a *determinant* of the matrix. It is defined as

$$|\mathbf{X}| = \sum \pm x_{1i_1} x_{2i_2} \dots x_{pi_m}$$

where the summation is over **all** permutations $(i, j, \dots, m)$ of the numbers $(1, 2, \dots, p)$ by taking into account the **sign rule**: summands with an even permutation get a $(+)$ whereas the ones with an odd permutation get a $(-)$ sign. It can be seen that this is equivalent to another recursive definition, namely:

- when $p = 1$ (scalar case) $\mathbf{X} = a$ is just a number and $|\mathbf{X}| = a$ in this case

- when $p = 2$ then $\begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix} = x_{11}x_{22} - x_{12}x_{21}$

- when $p = 3$ then the following rule applies:

$$\begin{vmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{vmatrix} = x_{11}x_{22}x_{33} + x_{12}x_{23}x_{31} + x_{21}x_{32}x_{13} - x_{31}x_{22}x_{13} - x_{11}x_{23}x_{32} - x_{12}x_{21}x_{33} \quad (1.8)$$

- recursively, for $\mathbf{X} \in \mathcal{M}_{(\mathbf{p},\mathbf{p})}$ we can define

$$|\mathbf{X}| = \sum_i (-1)^{i+j} x_{ij} |\mathbf{X}_{ij}| = \sum_j (-1)^{i+j} x_{ij} |\mathbf{X}_{ij}|$$

where $\mathbf{X}_{ij}$ denotes the matrix we get by deleting the i th row and j th column of $\mathbf{X}$.

Here we list some elementary properties of determinants that follow directly from the definition:

i) If one row or one column of the matrix contains zeros only then the value of the determinant is zero.

ii) $|\mathbf{X}'| = |\mathbf{X}|$

iii) If one row (or one column) of the matrix is modified by multiplying with a scalar c then so is the value of the determinant.

iv) $|c\mathbf{X}| = c^p |\mathbf{X}|$

v) If $\mathbf{X}, \mathbf{Y} \in \mathcal{M}_{p,p}$ then $|\mathbf{XY}| = |\mathbf{X}| \cdot |\mathbf{Y}|$

vi) If the matrix $\mathbf{X}$ is *diagonal* (i.e. all non-diagonal elements are zero) then $|\mathbf{X}| = \prod_{i=1}^p x_{ii}$. In particular, the **determinant of the identity matrix is always equal to one**.

Given that $|\mathbf{X}| \neq 0$ (or equivalently, if the matrix $\mathbf{X} \in \mathcal{M}_{p,p}$ is *nonsingular* then an **inverse** matrix $\mathbf{X}^{-1} \in \mathcal{M}_{p,p}$ can be defined that has to satisfy $\mathbf{X} \cdot \mathbf{X}^{-1} = \mathbf{I}_{p,p}$. It is easy to check that the inverse $\mathbf{X}^{-1}$ has as its (j, i) th entry $\frac{|\mathbf{X}_{ij}|}{|\mathbf{X}|} \cdot (-1)^{i+j}$.

Some elementary properties of inverses follow:

i) $\mathbf{X} \cdot \mathbf{X}^{-1} = \mathbf{X}^{-1} \cdot \mathbf{X} = \mathbf{I}$

ii) $(\mathbf{X}^{-1})' = (\mathbf{X}')^{-1}$

iii) $(\mathbf{X}\mathbf{Y})^{-1} = \mathbf{Y}^{-1} \cdot \mathbf{X}^{-1}$ when both $\mathbf{X}$ and $\mathbf{Y}$ are nonsingular square matrices of the same dimension.

iv) $|\mathbf{X}^{-1}| = |\mathbf{X}|^{-1}$

v) If $\mathbf{X}$ is diagonal and nonsingular then all its diagonal elements are nonzero and $\mathbf{X}^{-1}$ is again diagonal with diagonal elements equal to $\frac{1}{x_{ii}}, i = 1, 2, \dots, p$.

4.2.2. Rank

A set of vectors $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_k \in \mathbf{R}^n$ is *linearly dependent* if there exist k numbers $a_1, a_2, \dots, a_k$ **not all zero** such that

$$a_1 \mathbf{X}_1 + a_2 \mathbf{X}_2 + \dots + a_k \mathbf{X}_k = 0 \quad (1.9)$$

holds. Otherwise the vectors are *linearly independent*. In particular, for k linearly independent vectors the equality (1.9) would only be possible if **all** numbers $a_1, a_2, \dots, a_k$ were zero. **The row rank of a matrix is the maximum number of linearly independent row vectors.** The *column rank* is the rank of its set of column vectors. **It turns out that the row rank and the column rank of a matrix are always equal.** Thus the rank of a matrix $\mathbf{X}$ (denoted $rk(\mathbf{X})$) is either the row or the column rank. If $\mathbf{X} \in \mathcal{M}_{p,n}$ and $rk(\mathbf{X}) = \min(p, n)$ we say that the matrix is of full rank. In particular, a square matrix $\mathbf{A} \in \mathcal{M}_{p,p}$ is of full rank if $rk(\mathbf{A}) = p$. As is well known from the basic theorem of linear algebra *Theorem of Kronecker* this means also that $|\mathbf{A}| \neq 0$ when $\mathbf{A}$ is of full rank. Then the inverse of $\mathbf{A}$ will also exist. Let $\mathbf{B} \in \mathbf{R}^p$ be a given vector. Then the linear equation system $\mathbf{A}\mathbf{X} = \mathbf{B}$ has a unique solution $\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} \in \mathbf{R}^p$.

4.2.3. Traces and properties of traces

The trace of a square matrix $\mathbf{X} \in \mathcal{M}_{p,p}$ is denoted by $tr(\mathbf{X}) = \sum_{i=1}^p x_{ii}$. The following properties of traces are easy to obtain:

i) $tr(\mathbf{X} + \mathbf{Y}) = tr(\mathbf{X}) + tr(\mathbf{Y})$

ii) $tr(\mathbf{X}\mathbf{Y}) = tr(\mathbf{Y}\mathbf{X})$

iii) $tr(\mathbf{X}^{-1}\mathbf{Y}\mathbf{X}) = tr(\mathbf{Y})$

iv) If $\mathbf{A} \in \mathbf{R}^p$ and $\mathbf{X} \in \mathcal{M}_{p,p}$ then $\mathbf{A}'\mathbf{X}\mathbf{A} = tr(\mathbf{X}\mathbf{A}\mathbf{A}')$

4.2.4. Orthogonal matrices

A square matrix $\mathbf{X} \in \mathcal{M}_{p,p}$ is *orthogonal* if $\mathbf{X}\mathbf{X}' = \mathbf{X}'\mathbf{X} = \mathbf{I}_{p,p}$ holds. The following properties of orthogonal matrices are obvious:

i) $\mathbf{X}$ is of full rank ($rk(\mathbf{X}) = p$) and $\mathbf{X}^{-1} = \mathbf{X}'$

ii) the name *orthogonal* of the matrix originates from the fact that the scalar product of each two different column vectors equals zero. The same holds for the scalar product of each two different row vectors of the matrix. The norm of each column vector (or each row vector) is equal to one. These properties are equivalent to the definition.

iii) $|\mathbf{X}| = \pm 1$

4.2.5. Symmetric matrices. Eigenvalues and eigenvectors

If the matrix $\mathbf{X} \in \mathcal{M}_{p,p}$ has elements $x_{ij} = x_{ji}, i = 1, 2, \dots, p, j = 1, 2, \dots, p$ it is called a *symmetric matrix*.

For **any** square matrix $\mathbf{X} \in \mathcal{M}_{p,p}$ we can define its set of p *eigenvalues* which are the roots of the polynomial equation

$$f(\lambda) = |\mathbf{X} - \lambda \mathbf{I}| = 0 \quad (1.10)$$

Equation (1.10) is a polynomial equation of power p so it has exactly p roots. In general, some of them may be complex and some may coincide. Since the coefficients are real, if there is a complex root of (1.10) then also its complex conjugate must be a root of the same equation. Each of the above p roots is called *eigenvalue* of the matrix $\mathbf{X}$. Denote **any** such eigenvalue by λ^* . Obviously, the matrix $\mathbf{X} - \lambda^* \mathbf{I}$ is singular (its determinant is zero). Then, according to the Kronecker theorem, there exists a non-zero vector $\mathbf{y} \in \mathbf{R}^p$ such that $(\mathbf{X} - \lambda^* \mathbf{I})\mathbf{y} = \mathbf{0}, \mathbf{0} \in \mathbf{R}^p$. We call $\mathbf{y}$ an *eigenvector* of $\mathbf{X}$ that corresponds to the eigenvalue λ^* . Note that the eigenvector is not uniquely defined: $\mu \mathbf{y}$ for any real non-zero μ would also be an eigenvector corresponding to the same eigenvalue. Sparing some details of the derivation, we shall formulate the following basic result:

*When the matrix $\mathbf{X}$ is real symmetric then **all** of its p eigenvalues are **real**. If the eigenvalues are all different then all the p eigenvectors that correspond to them, are orthogonal (and hence form a basis in $\mathbf{R}^p$). These eigenvectors are also unique (up to the norming constant μ above). If some of the eigenvalues coincide then the eigenvectors corresponding to them are not necessarily unique but even in this case they can be chosen to be mutually orthogonal.*

For each of the p eigenvalues $\lambda_i, i = 1, 2, \dots, p$ of $\mathbf{X}$, denote its corresponding set of mutually orthogonal eigenvectors of *unit length* by $\mathbf{e}_i, i = 1, 2, \dots, p$, i.e.

$$\mathbf{X}\mathbf{e}_i = \lambda_i \mathbf{e}_i, i = 1, 2, \dots, p, \|\mathbf{e}_i\| = 1, \mathbf{e}_i' \mathbf{e}_j = 0, i \neq j$$

holds. Then it can be shown that the following decomposition (*spectral decomposition*) of any symmetric matrix $\mathbf{X} \in \mathcal{M}_{p,p}$ holds:

$$\mathbf{X} = \lambda_1 \mathbf{e}_1 \mathbf{e}_1' + \lambda_2 \mathbf{e}_2 \mathbf{e}_2' + \dots + \lambda_p \mathbf{e}_p \mathbf{e}_p' \quad (1.11)$$

The above decomposition is a very important analytical tool. One of its most widely used applications is for defining a square root of a symmetric positive definite matrix. A symmetric matrix $\mathbf{X} \in \mathcal{M}_{p,p}$ is *positive definite* if all of its eigenvalues are positive (it is called *non-negative definite* if all eigenvalues are ≥ 0). For a symmetric positive definite matrix we have all $\lambda_i, i = 1, 2, \dots, p$ to be positive in the spectral decomposition (1.11).

We can represent (1.11) in a matrix form $\mathbf{X} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ where $\mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_p \end{pmatrix}$ is

diagonal and $\mathbf{P} \in \mathcal{M}_{p,p}$ is an *orthogonal matrix* containing the p orthogonal eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_p$. But then

$$\mathbf{X}^{-1} = (\mathbf{P}')^{-1}\mathbf{\Lambda}^{-1}\mathbf{P}^{-1} = (\mathbf{P}^{-1})^{-1}\mathbf{\Lambda}^{-1}\mathbf{P}^{-1} = \mathbf{P}\mathbf{\Lambda}^{-1}\mathbf{P}' = \sum_{i=1}^p \frac{1}{\lambda_i} \mathbf{e}_i \mathbf{e}_i' \quad (1.12)$$

(i.e. inverting $\mathbf{X}$ is very easy if the spectral decomposition of $\mathbf{X}$ is known. Moreover we can define the *square root* of the symmetric non-negative definite matrix $\mathbf{X}$ in a natural way:

$$\mathbf{X}^{\frac{1}{2}} = \sum_{i=1}^p \sqrt{\lambda_i} \mathbf{e}_i \mathbf{e}_i' \quad (1.13)$$

The definition (1.13) makes sense since $\mathbf{X}^{\frac{1}{2}}\mathbf{X}^{\frac{1}{2}} = \mathbf{X}$ holds. Note that $\mathbf{X}^{\frac{1}{2}}$ is also symmetric and non-negative definite. Also $\mathbf{X}^{-\frac{1}{2}} = \sum_{i=1}^p \lambda_i^{-\frac{1}{2}} \mathbf{e}_i \mathbf{e}_i' = \mathbf{P}\mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{P}'$ can be defined where $\mathbf{\Lambda}^{-\frac{1}{2}}$ is a diagonal matrix with $\lambda_i^{-1/2}, i = 1, 2, \dots, p$ being its diagonal elements. These facts will be used essentially in the subsequent sections.

As an illustration of the usefulness of the spectral decomposition approach we shall show the following statement:

Exercise 1. If $\mathbf{X} \in \mathcal{M}_{p,p}$ is symmetric *positive definite* matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$ and associated eigenvectors of unit length $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_p$ then:

$$\max_{\mathbf{y} \neq \mathbf{0}} \frac{\mathbf{y}'\mathbf{X}\mathbf{y}}{\mathbf{y}'\mathbf{y}} = \lambda_1 \text{ attained when } \mathbf{y} = \mathbf{e}_1$$

$$\min_{\mathbf{y} \neq \mathbf{0}} \frac{\mathbf{y}'\mathbf{X}\mathbf{y}}{\mathbf{y}'\mathbf{y}} = \lambda_p \text{ attained when } \mathbf{y} = \mathbf{e}_p$$

Proof. Let $\mathbf{X} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ be the decomposition (1.11) for $\mathbf{X}$. Denote $\mathbf{z} = \mathbf{P}'\mathbf{y}$. Note that $\mathbf{y} \neq \mathbf{0}$ implies $\mathbf{z} \neq \mathbf{0}$. Thus

$$\begin{aligned} \frac{\mathbf{y}'\mathbf{X}\mathbf{y}}{\mathbf{y}'\mathbf{y}} &= \frac{\mathbf{y}'\mathbf{P}\mathbf{\Lambda}\mathbf{P}'\mathbf{y}}{\mathbf{y}'\mathbf{y}} = \\ \frac{\mathbf{z}'\mathbf{\Lambda}\mathbf{z}}{\mathbf{z}'\mathbf{z}} &= \frac{\sum_{i=1}^p \lambda_i \mathbf{z}_i^2}{\sum_{i=1}^p \mathbf{z}_i^2} \leq \lambda_1 \frac{\sum_{i=1}^p \mathbf{z}_i^2}{\sum_{i=1}^p \mathbf{z}_i^2} = \lambda_1 \end{aligned}$$

If we take $\mathbf{y} = \mathbf{e}_1$ then having in mind the structure of the matrix $\mathbf{P}$ we have $\mathbf{z} = \mathbf{P}'\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ \vdots \\ 0 \end{pmatrix}$

and for this choice of $\mathbf{y}$ also $\frac{\mathbf{z}'\mathbf{\Lambda}\mathbf{z}}{\mathbf{z}'\mathbf{z}} = \frac{\lambda_1}{1} = \lambda_1$. The first part of the exercise is shown. Similar arguments (just changing the sign of the inequality) apply to show the second part.

In addition, you can try to show that $\max_{Y \neq 0, Y \perp e_1} \frac{Y'XY}{Y'Y} = \lambda_2$ holds. HOW!

4.2.6. Orthogonal Projection

Orthogonal projection of any vector $Y \in R^n$ on the space $\mathcal{L}(X)$ spanned by the columns of the matrix $\mathbf{X} \in \mathcal{M}_{n,p}$ is a linear operation (WHY (?)). Hence the result is a vector $Z \in R^n$ that has the representation $Z = PY$ where the matrix $P \in \mathcal{M}_{n,n}$ is called (orthogonal) **projector**. Since $Z \in \mathcal{L}(X)$ (being a projection in this space), the projection of Z on $\mathcal{L}(X)$ is Z itself. Hence $PY = Z = PZ = PPY = P^2Y$ or $(P - P^2)Y = 0 \rightarrow P^2 = P$ (since $Y \in R^n$ is arbitrary). Therefore, P should be *idempotent*. Further $(Y - Z)'Z = 0$ or $Y'(P' - I)PY = 0$ for all $Y \rightarrow (P' - I)P = 0$ or $P'P = P$. Taking transposes, $P'P = P'$ or $P = P'$ that is, P is symmetrical. So, the orthogonal projector is a symmetric and idempotent matrix.

Vice versa, consider a symmetric and idempotent matrix P . Then if we take any $Y \in R^n$ then for $Z = PY \rightarrow PZ = P^2Y = PY \rightarrow P(Y - Z) = 0$ (and also $P'(Y - Z) = 0$ since $P = P'$.) Consider $\mathcal{L}(P)$ (the space generated by the rows/columns of P). Now: $Z = PY \rightarrow Z \in \mathcal{L}(P)$ and $P'(Y - Z) = 0$ means that $Y - Z$ is perpendicular to $\mathcal{L}(P)$. Hence PY is the projection of Y on $\mathcal{L}(P)$.

Hence, we have seen that $P \in \mathcal{M}_{n,n}$ is an orthogonal projection matrix if and only if it is a symmetric and idempotent matrix.

Also, if P is an orthogonal projection on a given linear space $\mathcal{M}$ of dimension $\dim(\mathcal{M})$ then $I - P$ an orthogonal projection on the orthocomplement of $\mathcal{M}$. It holds $rk(P) = \dim(\mathcal{M})$.

Further, it can be seen that the rank of an orthogonal projector is equal to the sum of its diagonal elements.

Finally, it can be shown that if the matrix $\mathbf{X}$ above has a full rank then the projector $P_{\mathcal{L}(X)} = X(X'X)^{-1}X'$. If the matrix $\mathbf{X}$ is not of full rank then the **generalized inverse** $(X'X)^-$ of $X'X$ can be defined instead. Note that the generalized inverse may not be uniquely defined but no matter which version of it has been chosen, the matrix $X(X'X)^-X'$ is uniquely defined and is the orthogonal projector on the space $\mathcal{L}(X)$ spanned by the columns of X also in cases when the rank of $\mathbf{X}$ is not full.